E-commerce Checkout Drop-off Analysis

SQL + Power BI Portfolio Project Report

Executive Summary

This project analyzes a large e-commerce event dataset to understand cart abandonment and checkout drop-offs. The workflow transforms event-level data into session-level funnel metrics, investigates behavioral drivers, and prioritizes product categories for business action.

Headline Metric	Result
Total sessions	9,244,422
Viewed-only sessions	8,344,836
Cart abandoned sessions	281,237
Cart -> purchase sessions	280,650
Purchase without cart sessions	337,699
Cart-to-purchase rate	49.95%
Cart abandonment rate	50.05%

1. Data and Methodology

The source is an event-level e-commerce dataset containing product views, cart events and purchases. PostgreSQL was used for data loading, validation, indexing, sessionization, funnel logic, behavioral segmentation and category aggregation. Power BI was connected to prepared reporting tables for dashboarding and stakeholder exploration.

Key modeling decision

Purchase events can appear in sessions without a recorded cart event, so a naive purchase/cart ratio produced a rate above 100%. The funnel was corrected by explicitly separating Viewed Only, Cart Abandoned, Cart -> Purchase, and Purchase Without Cart.

2. Funnel Performance

Among qualifying cart sessions, 280,650 converted and 281,237 abandoned. This produces a 49.95% cart-to-purchase rate and a 50.05% cart abandonment rate.

3. Behavioral Drivers

Cart size

Cart Size	Cart Sessions	Abandoned	Abandonment Rate
1 item	385,757	207,223	53.72%
2 items	106,960	46,470	43.45%
3 items	36,423	14,270	39.18%

Cart Size	Cart Sessions	Abandoned	Abandonment Rate
4-5 items	21,549	8,382	38.90%
6+ items	11,198	4,892	43.69%

Single-item carts have the highest abandonment rate (53.72%). This is an association and does not establish that cart size causes abandonment.

Product exploration

Product Views	Cart Sessions	Abandoned	Abandonment Rate
1 view	92,674	44,052	47.53%
2 views	119,695	52,423	43.80%
3 views	78,100	35,772	45.80%
4-5 views	94,442	46,835	49.59%
6-10 views	97,914	53,696	54.84%
11+ views	79,062	48,459	61.29%

Abandonment rises sharply among heavy product explorers: 54.84% for 6-10 views and 61.29% for 11+ views.

Repeated product views

Repeat View Group	Sessions	Abandoned	Abandonment Rate
No Repetition	120,735	58,875	48.76%
1 Repeat View	157,569	72,825	46.22%
2-3 Repeat Views	146,694	71,660	48.85%
4-5 Repeat Views	58,978	31,701	53.75%
6+ Repeat Views	76,720	45,346	59.11%

Session behavior

Abandoned sessions average 14.42 minutes and 6.63 product views, compared with 10.53 minutes and 5.17 product views for Cart -> Purchase sessions.

Time of day

Period	Cart Sessions	Abandoned	Abandonment Rate
Night (00-05)	105,062	51,656	49.17%
Morning (06-11)	231,867	109,887	47.39%
Afternoon (12-17)	177,043	93,992	53.09%
Evening (18-23)	47,915	25,702	53.64%

4. Category Opportunity Analysis

Category prioritization combines cart volume, abandonment and repeat-view friction so management does not simply chase the highest percentage.

Category	Cart	Abandoned	Abandon %	Repeat %	Priority
computers.desktop	2,278	1,383	60.71	38.24	Critical Opportunity
construction.tools.drill	2,354	1,415	60.11	38.49	Critical Opportunity
appliances.kitchen.refrigerators	8,386	4,869	58.06	35.40	Critical Opportunity
appliances.kitchen.kettle	2,155	1,219	56.57	36.57	Critical Opportunity
electronics.audio.headphone	32,253	17,749	55.03	21.50	High Volume + High Abandonment
electronics.smartphone	327,582	147,052	44.89	20.40	High Volume - Monitor

5. Business Recommendations

Simplify checkout for single-item carts

Test guest checkout, fewer steps, earlier shipping-cost visibility and express payment. Track single-item abandonment and checkout completion.

Support high-exploration shoppers

Test comparison tools, clearer specifications, reviews, delivery/return information and recently-viewed modules for sessions with 6+ product views.

Prioritize critical categories

Investigate pricing, content quality, delivery, stock, reviews and checkout friction first in categories combining volume, abandonment and repetition.

Investigate afternoon/evening friction

Segment by device, traffic source, site performance and customer type before testing time-specific interventions.

Segment cart recovery by intent

Differentiate high-intent abandoned carts from low-intent sessions and measure incremental recovery, revenue and margin.

6. Power BI Dashboard Structure

• Page 1 - Executive Overview: overall funnel, abandonment KPIs and category overview.

• Page 2 - Abandonment Drivers: cart size, product exploration, session behavior, time of day and repeat views.

• Page 3 - Category Opportunities: volume vs abandonment, abandoned-session volume, business priority and action table.

7. Recommended Experiment Framework

Observation -> Hypothesis -> Intervention -> A/B Test -> KPI -> Decision. Example: 11+ product views are associated with 61.29% abandonment -> hypothesis: comparison difficulty -> test product comparison -> measure cart conversion among high-exploration sessions -> roll out only if incremental impact is positive.

8. Limitations

• Observational data supports association, not causation.

• Explicit checkout-exit reasons, payment failures, shipping fees, device and traffic source are unavailable.

• Purchase-without-cart sessions indicate tracking or journey differences and must be handled separately.

• Some repeat-view Power BI reporting uses a pre-aggregated result for performance.

Conclusion

Checkout drop-off is not evenly distributed across customer behavior or product categories. The strongest opportunities are to reduce friction for single-item carts, assist high-exploration shoppers, and focus investigation on categories where abandonment coincides with meaningful volume and repeated product exploration. Recommendations should be validated through controlled experiments.